



## **ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)**

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

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# **Project Proposal**

CSE 4206 : Digital Logic Design Lab

## **Team Members**

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# Smart Parking System

## Motivation

Manual parking management wastes time and creates unnecessary congestion, especially in small lots, where drivers start picking the convenient spots which causes other drivers to park in awkward ways. A digital logic solution avoids this by tracking which spots are taken, the next best spot to use and managing entry control when the lot reaches capacity. The solution is built with only combinational and sequential IC's. This project exercises both data computation and state-holding control, which makes it a good project to fulfill the goals of this lab.

## Description

The system manages an 8-bay parking garage (slots 0–7). An automated test rig using a 74LS164 serial-in parallel-out (SIPO) shift register generates dynamic bay-status sequences (1 =vacant, 0 =occupied). Vehicle arrival and departure pulses are synchronized to the master clock to drive capacity tracking. An 8-bit bus feeds status data to priority allocation, duration counter arrays, dynamic billing logic, and automated barrier controllers.

## Core Features

- **Proximity-Based Allocation:** An 8-to-3 priority encoder (74LS148) selects the highest-priority vacant bay. Slots 0–1 serve as premium bays, slots 2–3 as accessible bays, and slots 4–7 as standard bays.
- **Capacity & Occupancy Tracking:** A 4-bit up/down counter (74LS193) maintains vacant bay counts, initialized at 8. A 4-bit 2's complement subtractor (74LS283) computes Occupied Slots = 8 - Vacant Slots.
- **Duration & Dynamic Billing:** Each occupied bay connects to an individual 4-bit duration counter running on an accelerated 2 Hz clock. A 74LS151 MUX auto-scanned by a 3-bit counter selects departing bay durations, feeding a 74LS283 arithmetic unit to calculate  $\text{Fee} = (\text{Duration} * \text{Base Rate}) + \text{Tier Surcharge}$ .

- **Automated Gate Control:** Synchronized entry pulses and a 74LS85 4-bit comparator authorize entry only when vacant bays remain ( $\text{Vacant} > 0$ ). A gate latch (74LS74) and down-timer (74LS191) open the barrier for a fixed duration, triggering a red alert when full.

## 1. Input (Automated Test Bench & Clock Network)

- **Automated Bay Occupancy Generator (74LS164 Shift Register):** Replaces manual toggles/switches. An 8-bit Serial-In Parallel-Out (SIPO) shift register continuously streams programmed logic patterns (1 = vacant, 0 = occupied) to simulate vehicle arrivals and departures across slots 0–7 without manual intervention.
- **Automated Event Pulse Generator:** Replaces manual push buttons. Synchronized clock pulses automatically generate arrival and departure trigger signals to increment/decrement the master capacity counter and trigger the barrier gate cleanly.

## 2. Output

- **7-Segment Displays:**
  - **Recommended Slot Display:** Outputs the 3-bit BCD value from the priority encoder indicating the lowest-numbered available bay.
  - **Vacant Capacity Display:** Indicates the real-time number of available spaces (0–8) from the capacity counter.
  - **Occupied Count Display:** Shows the total number of parked vehicles computed by the subtractor block.
  - **Total Checkout Fee Display:** Displays the dynamic fee for the bay currently selected by the auto-scanning MUX.
- **LED Indicators:**
  - **Green LED:** Indicates authorized entry (barrier open, vacant spots available).
  - **Red LED:** Indicates entry denied and full-lot alert ( $\text{Vacant} = 0$ ).

### 3. Adder / Subtractor / Dynamic Billing Arithmetic

- **4-bit Occupancy Subtractor (74LS283 via 2's Complement):** Takes the current available spaces from the capacity counter and subtracts them from total capacity (8) to yield occupied bays ( $\text{Occupied} = 8 - \text{Vacant}$ ), driving the occupied count display decoder.
- **Dynamic Fee Calculator (74LS283 Adder):** Calculates the total fee based on stay duration and bay tier:  $\text{Fee} = (\text{Duration} * \text{Base Rate}) + \text{Tier Surcharge}$ . EV/Premium slots (0–1) automatically add a flat surcharge via combinational adder logic.

### 4. Encoder / Decoder / MUX / DEMUX / Comparator

- **Priority Encoder (74LS148):** Evaluates the 8-bit bay status vector simultaneously. It bypasses occupied (0) signals and outputs the 3-bit binary address of the highest-priority vacant slot. Its Group Signal (GS) pin handles full-lot display blanking.
- **Checkout Selector MUX (74LS151):** An 8-to-1 multiplexer that selects the 4-bit duration counter output of the departing bay and routes it to the billing calculator.
- **Auto-Scan Binary Counter (74LS93):** Automatically cycles the select lines (S0, S1, S2) of the 74LS151 MUX every 1.5 seconds, sweeping across all 8 bays on the display without manual intervention.
- **BCD-to-7-Segment Decoders (74LS47 / 74LS48):** Translates binary outputs from the priority encoder, capacity counter, subtractor, and billing calculator into readable decimal formats on the displays.
- **Magnitude Comparator (74LS85):** Continuously tests whether available spots are greater than zero ( $\text{Vacant} > 0$ ). Authorizes entry pulses if true; triggers full-lot red alerts and inhibits entry if false ( $\text{Vacant} = 0$ ).

### 5. Register & Bus Network

- **8-bit Parallel Status Bus:** Directly routes the parallel outputs of the 74LS164 shift register across the priority encoder, clock-gating network, and duration counter array.

## 6. Counter

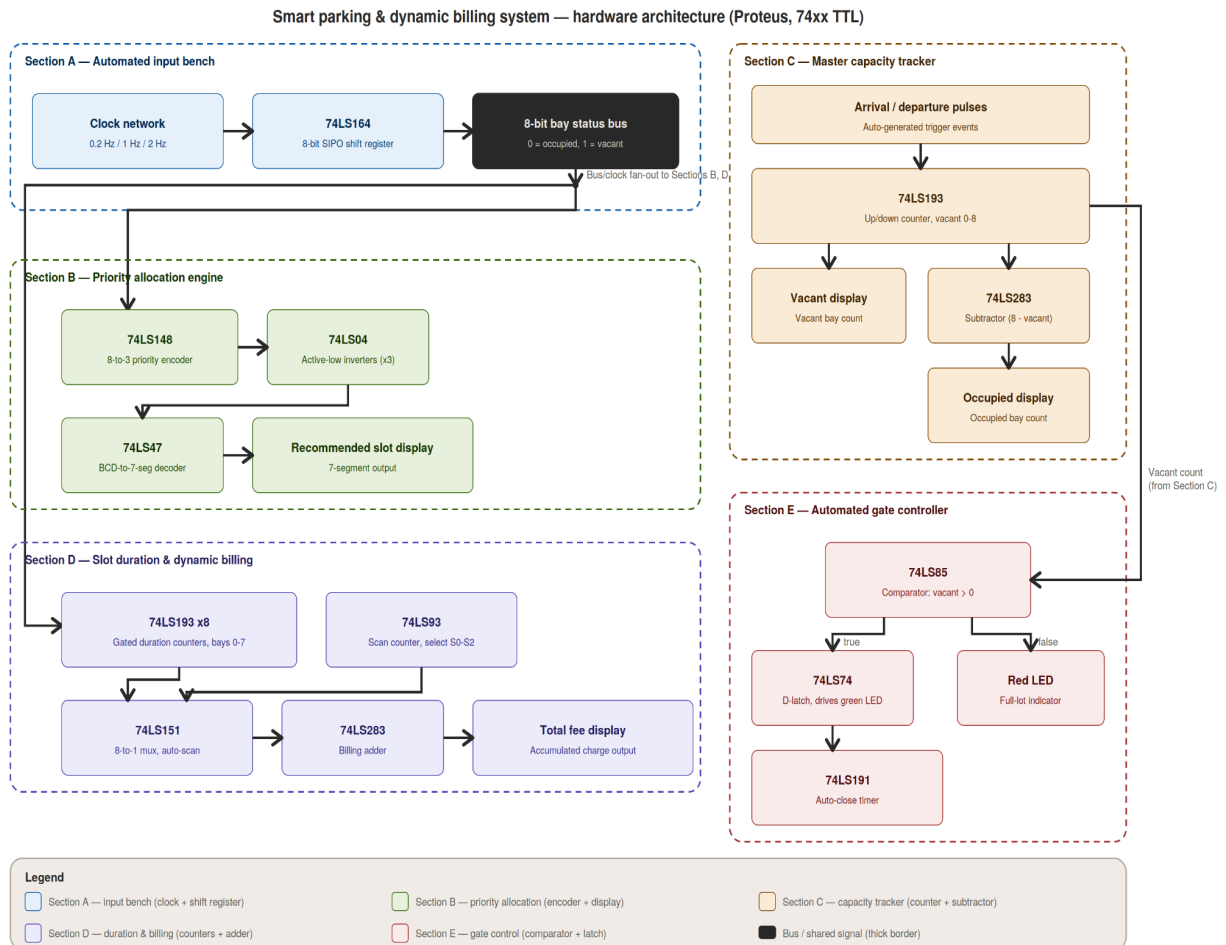
- **Up/Down Capacity Counter (74LS193):** Synchronous 4-bit counter initialized to 8. Decrement by automated entry pulses and increment by exit pulses to maintain the master vacant tally.
- **Slot Duration Counter Array (8x 74LS193):** Eight individual 4-bit counters (one per bay). Each counter receives a gated clock enable signal (CLK \* Bay Occupied) and tracks time units (0–15) while its respective bay is occupied. Driven by an accelerated 2 Hz master clock so full test cycles complete in under 5 seconds.
- **Gate Timer Down-Counter (74LS191):** Secondary down-counter that holds the virtual barrier open upon authorized entry and counts down to zero to automatically trigger gate closure.

## 7. Flip-Flop & Control Logic

- **Gate Control Latch (74LS74 D Flip-Flop):** Receives the entry-authorized signal from the comparator, latching HIGH to lift the barrier and activate the Green LED until the 74LS191 gate timer reaches zero and resets it to LOW.

| Member                     | Assigned Sub-module | Detailed Work Breakdown & IC Implementation                                                                 |
|----------------------------|---------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Md. Fahad Rahman</b>    | Automated Test Rig  | 74LS164 shift register sequence generator, 74LS93 auto-scan counter, and master clock distribution network. |
| <b>Mir M. Azmain Ahnaf</b> | Priority Allocation | 74LS148 priority encoder, active-low signal inversion logic, 74LS47 decoder, and full-lot blanking circuit. |
| <b>Eftakha Al Sabbir</b>   | Capacity Tracker    | 74LS193 up/down capacity counter, pulse steering, and 74LS283 2's complement occupied bay subtractor.       |
| <b>Waseef Munir</b>        | Duration Timers     | 8x 4-bit duration counters array with gated clock enable logic.                                             |
| <b>Hasnat Farhan Ahmed</b> | Billing Calculator  | 74LS151 8-to-1 checkout MUX routing, rate multiplier, and 74LS283 fee calculation arithmetic unit.          |
| <b>Jamil Hasan Riju</b>    | Gate Controller     | 74LS85 comparator, 74LS74 barrier D-latch, 74LS191 gate down-timer, LED indicators, and 7-segment displays. |

# Project Diagram



## Feedback

(To be provided by course teacher)